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MILITARY STANDARD

BALLISTIC NOMENCLATURE ROCKET STATIC TESTS

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Supply and Logistics
Ballistic Nomenclature Rocket Static Tests
MIL–STD–292C 24 FEBRUARY 1961

1. This standard has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force, effective 24 February 1961.

2. In accordance with established procedure, the Ordnance Corps, Bureau of Naval Weapons and Air Force have been designated as Army-Navy-Air Force custodians of this standard.

3. Recommended corrections, additions, or deletions should be addressed to the Standardization Division, Armed Forces Supply Support Center, Washington 25, D. C.

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FOREWORD

Automatic data-reduction equipment is proving to be more consistent and more accurate than manual assessment and easily justified costwise in production testing. It is possible at this time to determine delay times, integrals of pressure and thrust traces, the times over which the integrals are computed, and the maximum values of pressure or thrust traces, and to present this information in numerical form immediately at the completion of the static firing. This means that the only manual effort is in the recording and tabulating of the final results. Acceptance or rejection of the lot, or testing of an additional sample, can be determined immediately without the delay inherent in manual assessment.

There are, however, in effect certain techniques used in reducing ballistic data by manual assessment methods which cannot easily be performed by automatic machines. In some cases it would require a memory circuit from which the data could be resolved at a later time and in other cases the machine would be required to draw tangents and bisect angles; the development costs of such a machine would be prohibitive. This system provides alternates for the measurement of these parameters and the alternates should produce better control of the end product through the use of fixed threshold levels for a given rocket.

This standard is a guide by which the naming of ballistic parameters and symbols for them can follow a common pattern, and when used by all rocket facilities, provide a dictionary for static testing terminology.

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Figure 1. Graphic illustration of the use of standard symbols. 10

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1. SCOPE

1.1 SCOPE. This standard designates names and associated terminology for use when referring to rocket performance data obtained from static test firings. The main objective is to define acceptance parameters in such a manner that they can be measured by the latest type of automatic data-reducing instrumentation and yet provide an equivalent to existing parameters which cannot be reduced automatically.

1.2 APPLICATION. The terms and symbols defined herein are prescribed for use, where applicable, in specifications, standards, drawings, technical manuals, in contractual, inspection, and similar documents, and in engineering evaluation reports.

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2. REFERENCED DOCUMENTS

(Not Applicable)

3. DEFINITIONS

3.1 TIME.

3.1.1 *General.* The time measurements defined in this system are based on the use of fixed threshold levels. However, Ignition Delay Time (the time from zero time to the point on the pressure trace where the pressure has risen to 10 percent of its maximum value) and Action Time (the time from 10 percent of the maximum pressure on the rise portion of the trace to the same level on the decay portion of the trace) are good descriptive terms. Also, the desire to measure the time from Zero Time to some fixed level on a ballistic trace and the time from one point on the rise portion of the trace to the corresponding point on the decay portion of the trace still exists; therefore, these terms have been incorporated into this system in modified forms. As t_a and t_d occur in this system, they will always have a numeral associated with the subscript which will indicate the magnitude of the threshold level (see 3.1.3.3).

At the present time most measurements are computed from the pressure trace. The desire to base time requirements on the thrust or other traces exists now and may soon become a reality on certain units. The symbol p or f (or other) may be used as an additional final subscript to define the origin of the subscripts which precede (see 3.1.3.3).

3.1.2 *Rules for time names and symbols.*

3.1.2.1 The t shall be the symbol used to designate time measurements, T being reserved for temperature.

3.1.2.2 Seconds shall be used as the units for time measurements.

3.1.2.3 Zero time is the time at which voltage is applied to the igniter circuit.

3.1.3 *Subscripts.*

3.1.3.1 Parentheses ( ) placed about a subscript which is a pressure or thrust symbol or other with its own subscript, shall denote the time to that pressure or thrust or other measurement measured from Zero Time (see 3.5.1). This does not apply to measurements which are made from Zero Time by definition; e.g., t_d (see 3.1.4.2).

3.1.3.2 Subscripts without parentheses shall denote time measurements made between points on the ballistic record (see 3.1.4.6).

3.1.3.3 A numeral subscript placed after a letter subscript indicates a fixed pressure or thrust threshold level. The number indicates the threshold level divided by 100. If a pressure threshold is used to establish time measurements, then a final subscript p is added; if a thrust threshold level is used to establish time measurements, then a final subscript f is added, etc. (see 3.1.4.6 and 3.1.4.7). A threshold level of 100 psig or 100 lbf is employed in the examples below.

3.1.3.4 For a given rocket, if delay times and integral times are based on a pressure threshold level, then these times are used in reducing the thrust-time trace. If delay times and integral times are based on a thrust threshold level, then these times are used in reducing the pressure-time trace.

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3.1.4 *Specific time definitions.*

	Symbol	*Name*	*Definition*
3.1.4.1	t_o	Zero Time.	Application of firing voltage to the igniter circuit.
3.1.4.2	t_d	Ignition delay time.[1]	The time from Zero Time to the point on the pressure or thrust trace where the pressure or thrust has risen to 10 percent of its maximum value.
3.1.4.3	t_{d1p}	100 psig delay time.	The time from Zero Time to the initial point on the pressure trace where the pressure has reached a sustained value in excess of 100 psig.
3.1.4.4	t_{d1f}	100 pound delay time.	The time from Zero Time to the initial point on the thrust trace where the thrust has reached a sustained value in excess of 100 pounds.
3.1.4.5	t_a	Action time.[2]	The time from 10 percent of the maximum pressure or thrust on the rise portion of the trace to the same level on the decay portion of the trace.
3.1.4.6	t_{a1p}	100 psig action time.	The time between the initial 100 psig level and the final 100 psig level on the pressure trace.
3.1.4.7	t_{a1f}	100 pound action time.	The time between the initial 100 pound level and the final 100 pound level on the thrust trace.

[1] Since this value cannot be measured by automatic equipment as at the time of firing, it should be replaced where possible by a term such as t_{d1p}.
[2] Since this value cannot be measured by automatic equipment at the time of firing, it should be replaced where possible by a term such as t_{a1p}.

3.2 PRESSURE

3.2.1 *General.* Two types of pressures may be indicated: specific point pressures (such as maximum pressure) and threshold pressures. Threshold pressures are indicated using subscripts to show the source of the threshold level (see 3.2.3.1).

3.2.2 *Rules for pressure names and symbols.*

3.2.2.1 P shall be used to denote pressure measured from a ballistic record.

3.2.2.2 Pressure shall be expressed as gage pressure and so indicated by psig.

3.2.2.2.1 Absolute pressure when necessary shall have the units shown as psia.

3.2.3 *Subscripts.*

3.2.3.1 A numeral subscript placed after a letter subscript on the symbol P or $\overline{P}$ indicates a fixed pressure or thrust threshold level. The number indicates the threshold level divided by 100. If a pressure threshold is used the final subscript p is added: if a thrust threshold is used the final subscript f is added. A threshold level of 100 psig is used in example 3.2.4.3 below.

3.2.3.2 A numerical subscript placed between the symbol P and its letter subscript t indicates the time in seconds at which the pressure measurement was taken after Zero Time (see 3.2.4.4).

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3.2.4 *Specific pressure definitions.*

	Symbol	*Name*	*Definition*
3.2.4.1	P_{max}	Maximum pressure.	The maximum pressure value that occurs at any one point on the pressure-time trace, excluding ignition shocks of an instantaneous nature.
3.2.4.2	P_{min}	Minimum pressure.	The minimum pressure that occurs between two maximum points such as in a saddle-type pressure-time trace.
3.2.4.3	$\overline{P}_{alp}$	Average between 100 psig levels.	The average pressure between the initial 100 psig level and the final 100 psig level on the pressure trace.
3.2.4.4	$P_{0.5t}$	Pressure at 0.5 second.	The value on the pressure trace at a time of 0.5 second after Zero Time. The numeral may be changed to fit the application.
3.2.4.5	P_{mi}	Maximum initial pressure.	The initial maximum pressure which occurs on the pressure-time trace.
3.2.4.6	P_{mf}	Maximum final pressure.[3]	The final maximum pressure which occurs on the pressure-time trace.

[3] The subscript "f" is used on this symbol to denote "final" since no ambiguity between "final" and "force" exists in this case.

3.3 THRUST.

3.3.1 *General.* Two types of thrusts may be indicated: specific-point thrusts (such as maximum thrust) and threshold thrusts. Threshold thrusts are indicated using subscripts to show the source of the threshold level (see 3.3.3.1).

3.3.2 *Rules for thrust names and symbols.*

3.3.2.1 F shall be the symbol for thrust or force measured from a static test firing.

3.3.2.2 Pounds force (abbreviated lbf) shall be the units in which the thrust is expressed.

3.3.3 *Subscripts.*

3.3.3.1 A numeral subscript placed after a letter subscript on the symbol F or $\overline{F}$ indicates a fixed pressure or thrust threshold level. The number indicates the threshold level divided by 100. If a pressure threshold is used the final subscript p is added; if a thrust threshold is used the final subscript f is added. A threshold level of 100 psig is used in example 3.3.4.3 below.

3.3.3.2 A numeral subscript placed between the symbol F and its letter subscript t indicates the time in seconds at which the thrust measurement was taken after Zero Time (see 3.3.4.4).

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3.3.4 *Specific thrust definitions.*

	Symbol	*Name*	*Definition*
3.3.4.1	F_{max}	Maximum thrust.	The maximum thrust value that occurs at any point on the thrust time trace. This value should not include impacts resulting from the resonant frequency of the static thrust system, unless the test system has a resonant frequency comparable to that of a system in which the rocket will be used.
3.3.4.2	F_{min}	Minimum thrust.	The minimum thrust that occurs between two maximum points such as in a saddle type thrust-time trace.
3.3.4.3	$\overline{F}_{alp}$	Average thrust between 100 psig levels.	The average thrust measured during the time when the pressure trace is above the level of 100 psig.
3.3.4.4	$F_{0.5t}$	Thrust at 0.5 second.	The value of thrust at a time of 0.5 second after Zero Time. The numeral may be changed to fit the application.
3.3.4.5	F_{mi}	Maximum initial thrust.	The initial maximum thrust which occurs on the thrust-time trace.
3.3.4.6	F_{mf}	Maximum final thrust.[4]	The final maximum thrust which occurs on the thrust-time trace.

[4] The subscript "*f*" is used on this symbol to denote "final" since no ambiguity exists between "final" and "force" in this case.

3.4 IMPULSE.

3.4.1 *Rules for impulse names and symbols.*

3.4.1.1 "I" shall be the symbol for impulse, the integral of force with respect to time between defined time limits (see 3.4.2.1).

3.4.1.2 Pound-force-seconds (abbreviated lbf-sec) shall be the units in which impulse is expressed.

3.4.1.3 I_{sm} shall be the symbol for motor specific impulse, the ratio of the impulse to the combined mass of the motor and propellant (see 3.4.2.2).

3.4.1.4 I_{sp} shall be the symbol for propellant specific impulse, the ratio of the impulse to the burned propellant mass alone (see 3.4.2.3).

3.4.1.5 Pound-force-seconds per pound mass (abbreviated lbf-sec/lbm) shall be the units in which specific impulse is expressed.

3.4.1.6 Specific impulse nomenclature shall indicate the impulse used and the mass to avoid misunderstandings.

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3.4.2 *Specific impulse definitions.*

	Symbol	*Name*	*Definition*
3.4.2.1	I	Impulse.	$\int_{t_1}^{t_2} F\,dt$
3.4.2.2	I_{sm}	Motor specific impulse (measured).	$\frac{1}{m_m}\int_{t_1}^{t_2} F\,dt$ m_m = total mass of loaded motor
3.4.2.3	I_{sp}	Propellant specific impulse (measured).	$\frac{1}{m_p - m_s}\int_{t_1}^{t_2} F\,dt$ m_s = mass of slivers m_p = mass of propellant
3.4.2.4	$I_{0.25t}$	Quarter-second impulse.	The impulse developed in the first quarter of a second after Zero Time. The numeral may be changed to fit the application.
3.4.2.5	$I°_{sp}$ or $I°_{sm}$	Calculated (not measured) specific impulse.	

3.5 SPECIALIZED PARAMETERS. Listed as follows are the specialized parameters which have been derived from the symbols listed above. These parameters serve to illustrate the versatility and flexibility of the system of ballistic nomenclature.

	Symbol	*Name*	*Definition*
3.5.1	$t_{(Pme)}$	Time to maximum erosion pressure.	The time from Zero Time to the point on the pressure trace where the maximum pressure due to erosion occurs (see 3.5.5)
3.5.2	$t_{(Pmin)}$	Time to minimum pressure.	The time from Zero Time to the point on the pressure trace where a minimum pressure occurs between two maximum pressures.
3.5.3	t_u	Useful time.	The time from some fixed level on the initial portion of pressure or thrust trace to some higher fixed level on the decay portion of the same trace. In actual use, numerals must be associated with the subscript "u" to denote the levels. Example: $t_{u1\text{-}9f}$ is the useful time from 100 pounds of thrust on the initial portion of the thrust trace to the point where the thrust has decayed to 900 pounds.
3.5.4	t_i	Ignition time.	The time from some fixed level on the initial portion of the pressure or thrust trace to some higher level on the initial portion of the same trace.

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	Symbol	*Name*	*Definition*
3.5.5	P_{me}	Maximum erosion pressure.	The maximum pressure which occurs early on the pressure-time trace which is produced by an increased burning rate of the propellant due to erosive burning during the period of low port area.
3.5.6	P_{mp}	Maximum progressivity pressure.[5]	The maximum pressure which occurs late in the pressure-time trace and results from an increased burning surface of the propellant.
3.5.7	F_{me}	Maximum erosion thrust.	The maximum thrust which occurs early on the thrust-time trace and which is due to erosion.
3.5.8	F_{mp}	Maximum progressivity thrust.[5]	The maximum thrust which occurs late in the thrust-time trace and is due to progressivity.

[5] The subscript p is used on this symbol to denote "progressivity" since no ambiguity between "progressivity" and "pressures" exists in this case.

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4. GENERAL REQUIREMENTS

4.1 SUBSCRIPTS.

4.1.1 All subscripts shall be written on the same level, which shall be one level below the main symbol (e.g., t_{a1f}), except where a subscript letter itself has a subscript (e.g., t $(_{pme})$.

4.1.2 Subscripts shall reflect the name given to the measurement (e.g., t_a action time).

4.2 SUPERSCRIPTS.

4.2.1 A superscript (°) on the main symbol shall be used to indicate a calculated value; i.e., one not measured in the static test (e.g., I°).

4.2.2 A superscript ——— (bar) shall be used over the main symbol to denote an average value, (e.g., $\overline{P}$).

4.2.3. Figure 1 on page 10 graphically illustrates the majority of standard symbols used.

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5. DETAIL REQUIREMENTS
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Copies of this standard for military use may be obtained as indicated in the foreword to the Index of Military Specifications and Standards.

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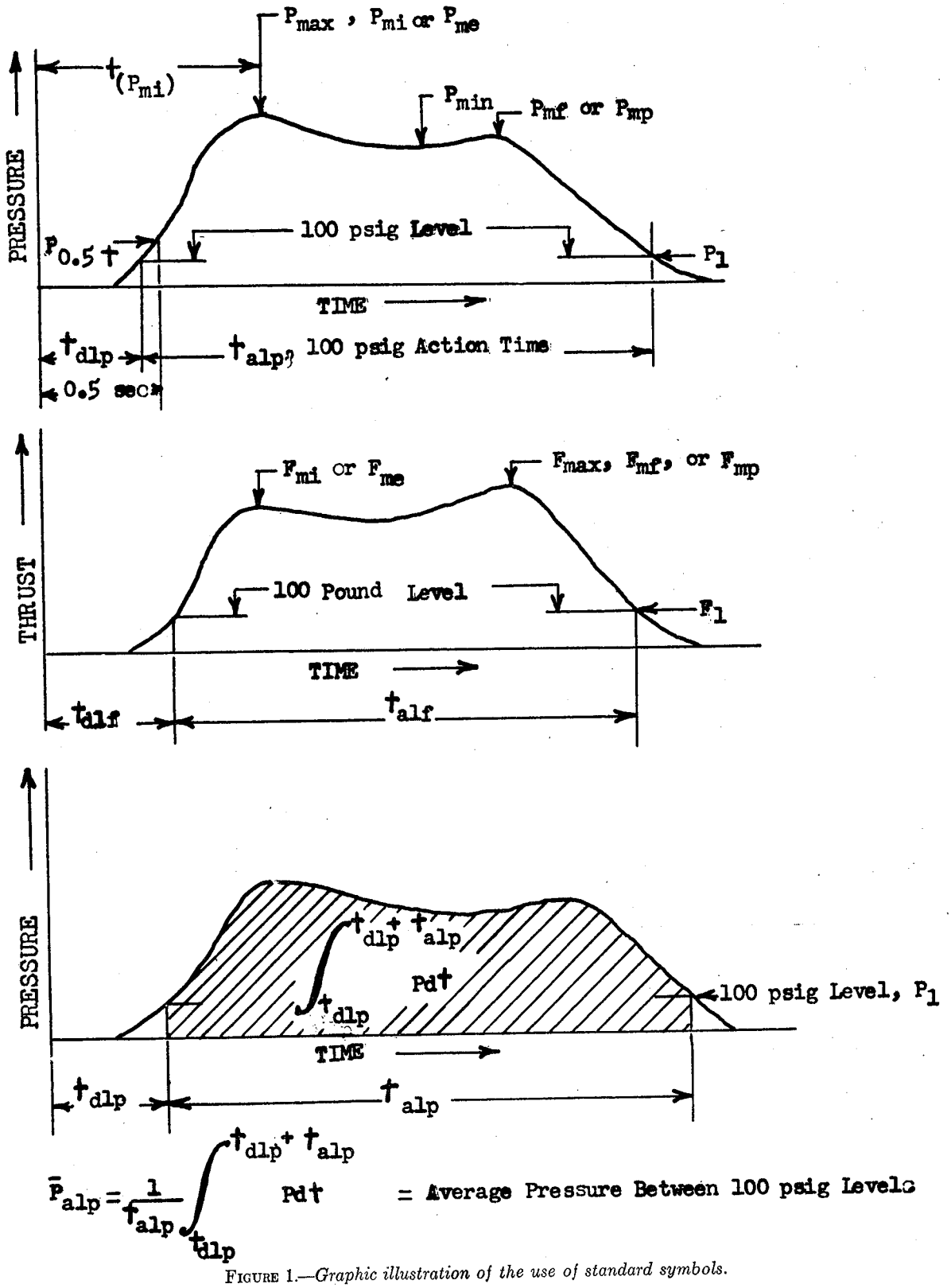


FIGURE 1.—*Graphic illustration of the use of standard symbols.*

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